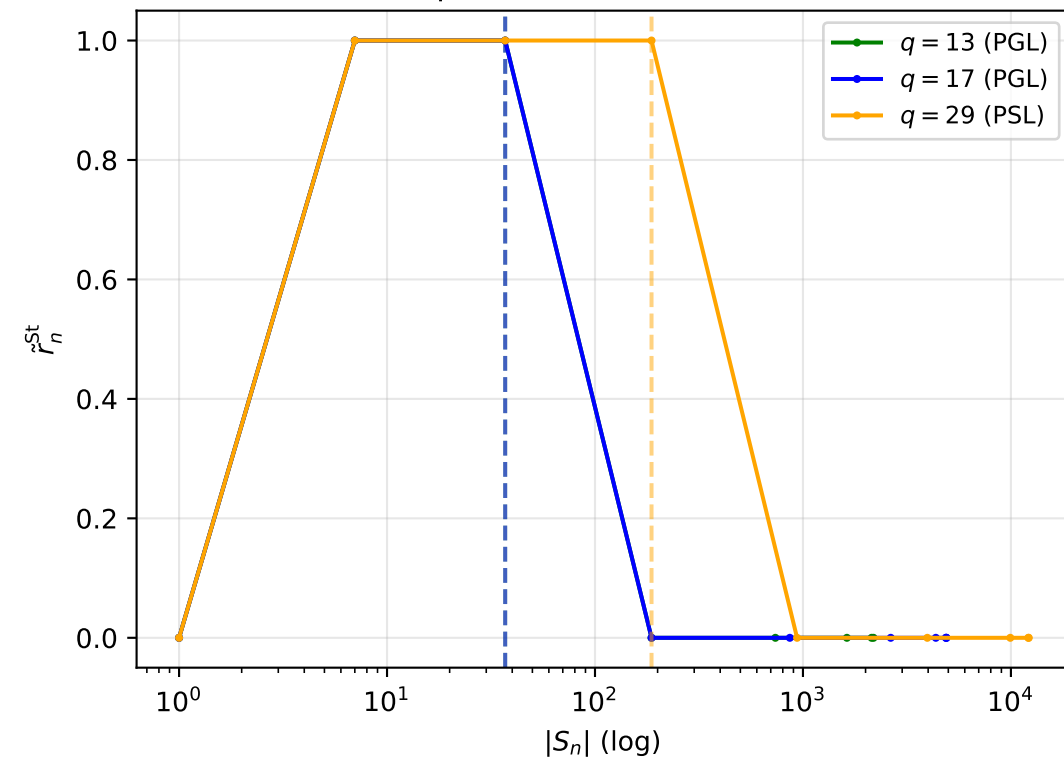
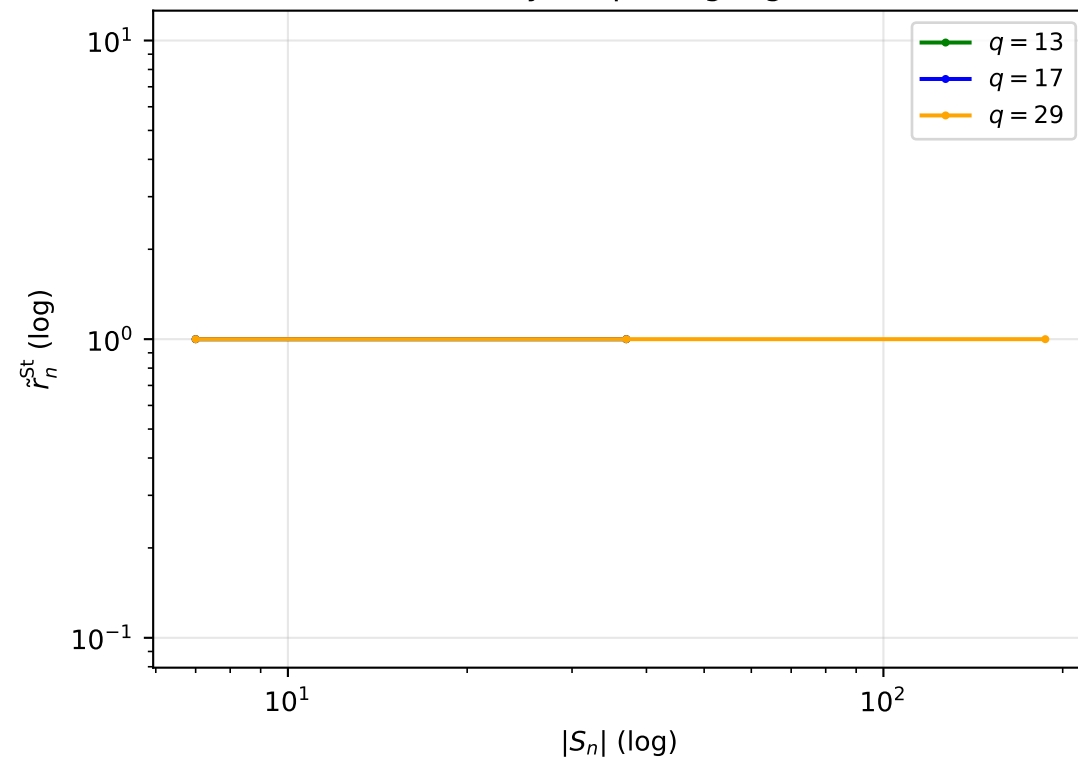


Steinberg $\tilde{\sigma}_u \odot \tilde{\sigma}_s$ pre-saturation law -- $X^{5,q}$, $q \in 13, 17, 29$, shell-layered

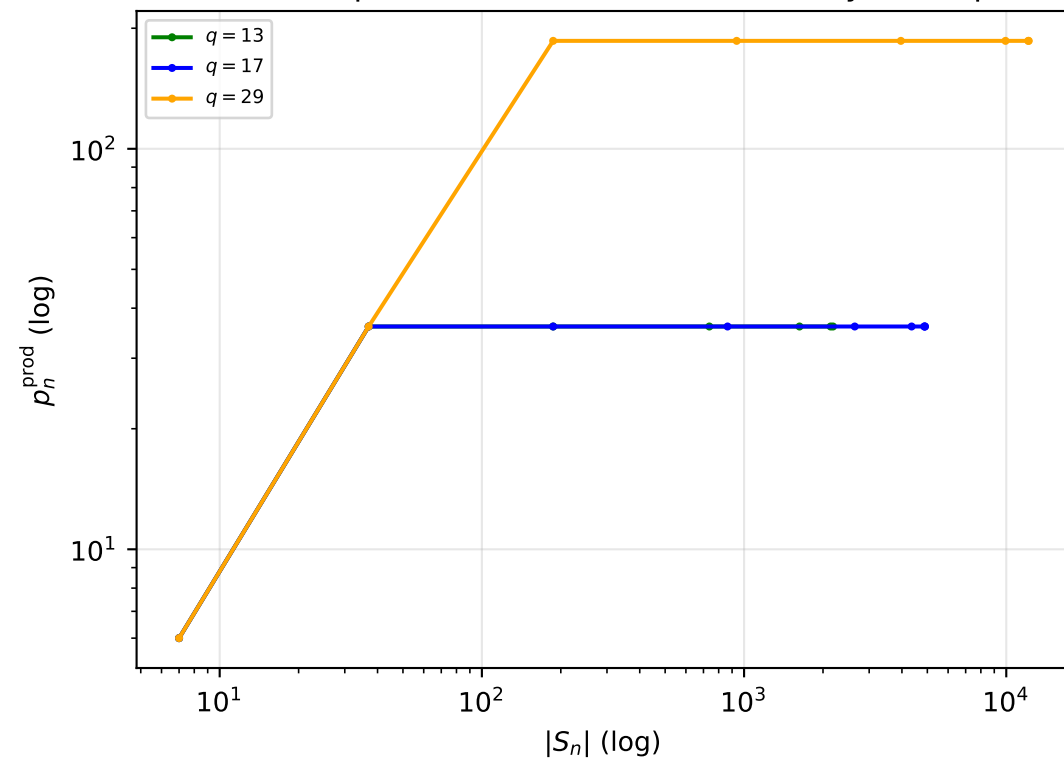
Trace-productive frontier fraction



Decay shape (log-log)



Cumulative trace-productive front (reference only; no exponent fit)



Summary

PARAMETER TABLE (shell-layered, verified group)

$q = 13$ (PGL): $|G| = 2184$, $|S^*| = 37$, $|S^*|/|G| = 0.01694$
 $q = 17$ (PGL): $|G| = 4896$, $|S^*| = 37$, $|S^*|/|G| = 0.00756$
 $q = 29$ (PSL): $|G| = 12180$, $|S^*| = 187$, $|S^*|/|G| = 0.01535$

Usable pre-saturation windows for a beta_prod fit: 0/3
 Saturation occurs inside the local tree-like regime
 shared by every tested q (shell sizes 1,6,30,150,... match
 the free 6-regular tree exactly before any collision).